HOPPER INFORMATION SERIES

05 / BINARY COMPUTER VISION

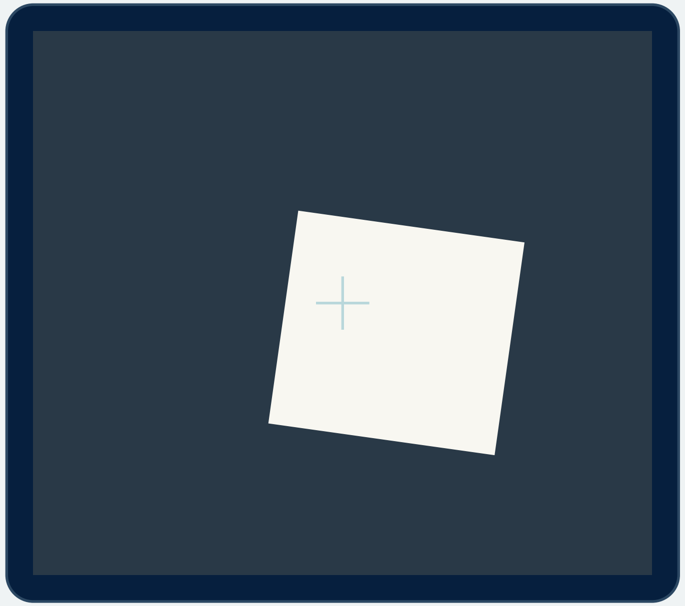
Thresholding with Hopper

Turn camera brightness into a reliable two-color signal for a white-paper mission.

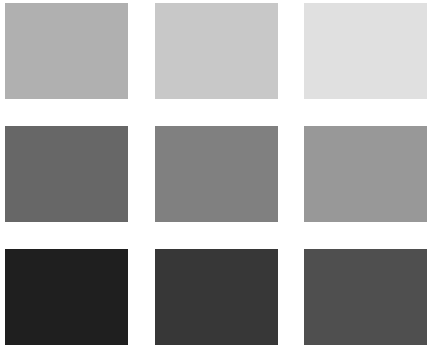
WHITE + BLACK | 0-100% THRESHOLD | DOWNWARD CAMERA

BINARY VISION PIPELINE

One camera frame becomes two colors

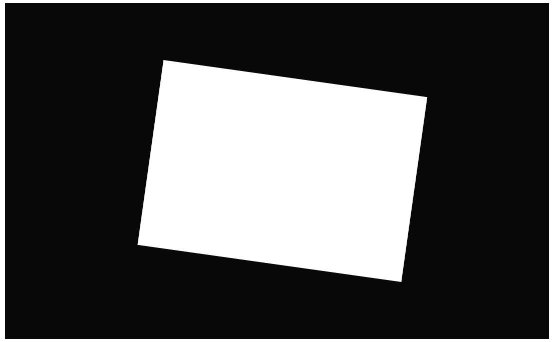


GRAYSCALE



brightness I(x,y)

BINARY OUTPUT

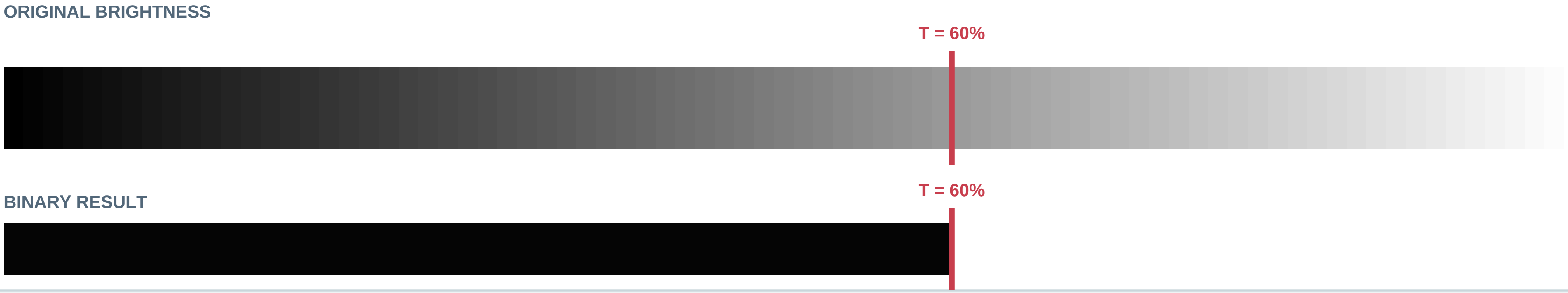


0 or 1

CAMERA -> LUMINANCE -> THRESHOLD RULE -> WHITE / BLACK

BRIGHTNESS THRESHOLD

The decision rule is simple and inspectable




```
B(x,y) = 1  if  I(x,y) >= T
otherwise B(x,y) = 0
```

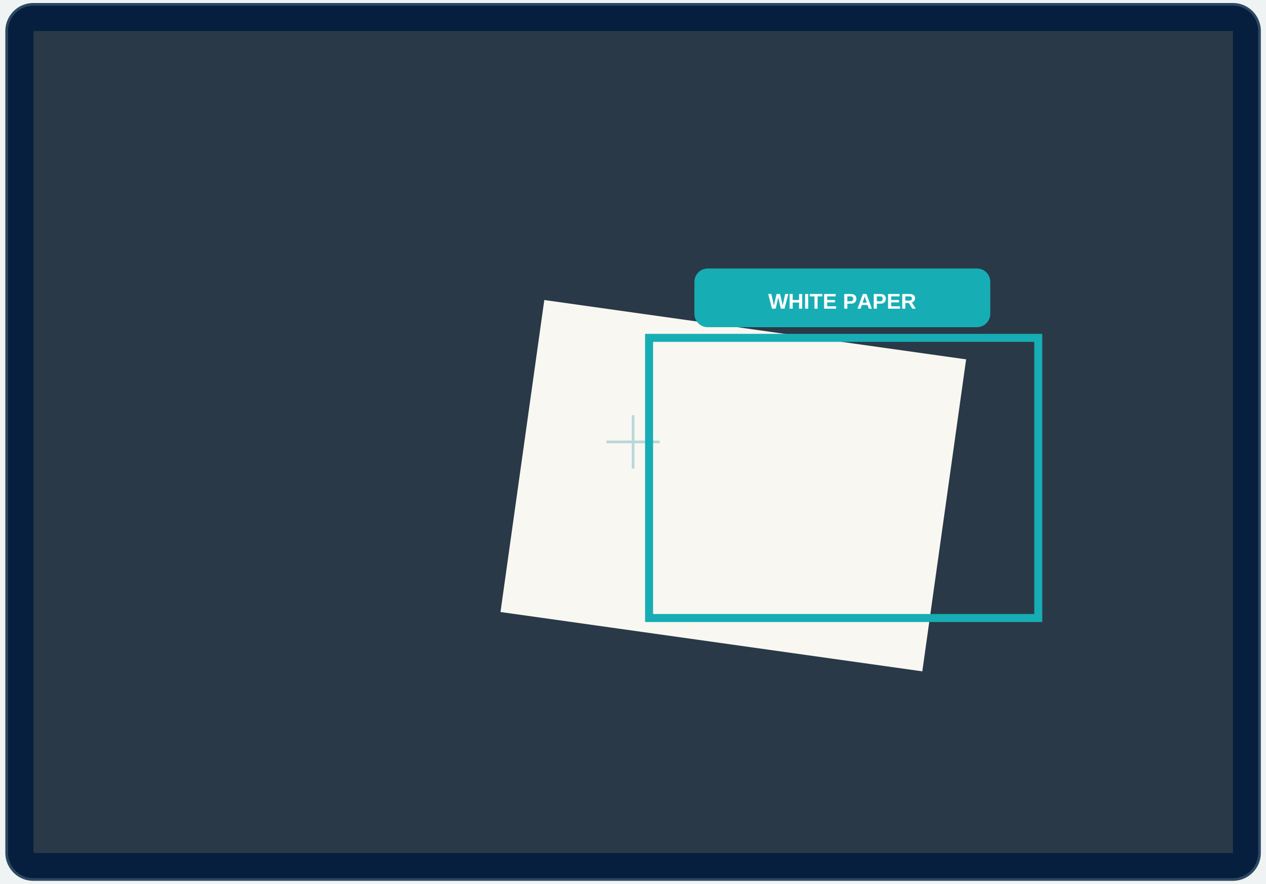
invert swaps 0 and 1

WHAT THE SCAN RETURNS

- whiteCoverage and blackCoverage in percent.
- centerWhite Boolean at the reticle pixel.
- frame width/height and the binary image.

DOWNWARD-CAMERA EXPERIMENT

A white sheet can become a visual landing cue



CALIBRATE BEFORE FLIGHT

- Place the real paper on the real floor.
- Use the exact classroom lighting.
- Adjust T until paper stays white and floor stays black.
- Record the chosen threshold.

CHOOSE A DECISION

- Coverage: enough of the frame is paper.
- Center pixel: reticle has crossed onto paper.
- Combine repeated checks for a safer decision.
- Then explicitly command land.

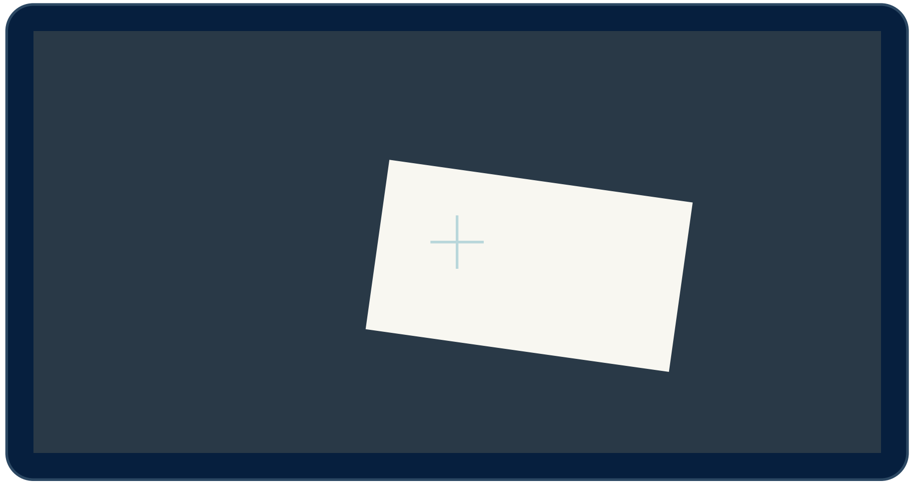
Thresholding detects brightness, not the semantic concept 'paper'.

COVERAGE VS CENTER PIXEL

Two useful sensing strategies

FRAME COVERAGE

Ask: what percentage of all pixels are white?

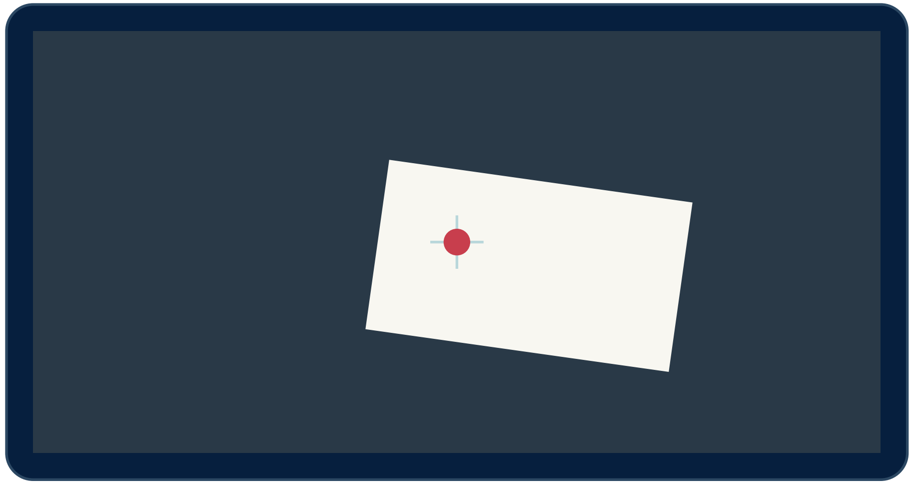


```
whiteCoverage >= 18%
```

Good for: area entered, target fills view

CENTER PIXEL

Ask: is the single reticle pixel white?



```
centerWhite === true
```

Good for: center over target, line crossing

A CONSERVATIVE MISSION

Search, verify, then land

1 Take off
Stabilize over clear floor.

→

2 Move
Small search-pattern segment.

→

3 Scan
Fresh binary frame.

→

4 Verify
Center or coverage true twice.

→

5 Land
Explicit landing command.

SEARCH CHECK

```
const hit = await
  vision.seesBinary("white", 60, false, 18);
if (hit) {
  await drone.hover();
  await drone.land();
}
```

SAFER THAN ONE CHECK

- Move in small increments.
- Pause for exposure and motion to settle.
- Require repeated consistent observations.
- Keep a timeout and fallback land.

FAILURE MODES

Lighting is part of the experiment

SHADOW

A shadow can turn white paper gray or black.

REFLECTION

Glare can turn dark floor pixels bright.

AUTO-EXPOSURE

The same paper may shift brightness as the view changes.

MOTION BLUR

Fast movement mixes paper and floor at the edge.

FLOOR TEXTURE

Light markings can resemble the target.

TARGET SIZE

A small target may never reach the coverage threshold.

SOURCES AND FURTHER READING

Hopper Studio - threshold implementation and scan behavior
lib/vision.ts and README.md in the Hopper Studio project

FTW Robotics - indoor lighting and floor guidance
https://ftw-robotics.ai/hopper